

Question				Marking details	Marks Available				
					AO1	AO2	AO3	Total	Maths * Prac **
2	(a)			Allowing {sodium <u>ions</u> / Na^+} to {rapidly diffuse/ flood/ rapid influx} <u>into</u> {neurone/ axon/cell} (1) Brings about depolarisation of the axon / inside of neurone becomes positive with respect to outside / generation of action potential (1)	2			2	
	(b)			A. {sodium <u>ions</u> / Na^+} can no longer enter/ fewer sodium ions enter (the axon)(1) B. Threshold is not reached(1) C. There is no action potential / no depolarisation (1) D. Along <u>sensory</u> neurone (to brain) (1)	2	2		4	
	(c)			Correct answer = $21 = 2$ marks Allow 1 mark for $7 \times \frac{60 \times 0.5}{10}$ OR $7 \times 6 \times 0.5$ OR $7 \times \frac{60 \times 1}{2 \times 10}$ OR $7 \times 6 \times \frac{1}{2}$		2		2	2
	(d)			A. Calcium ions {do not <u>diffuse</u>/ less <u>diffuse</u>} into the {synaptic knob/ presynaptic neurone/ across the presynaptic membrane}(1) B. Prevent (synaptic) vesicles from fusing with presynaptic membrane (1) C. Prevents {neurotransmitter/ acetylcholine} {release into the (synaptic) cleft/ synapse/ exocytosis} (1)		3		3	
				Question 2 total	4	7	0	11	2 0